

Statistical-mechanical characterization of memory and quench response in state-separated LLM-agent networks

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Abstract

We treat a network of state-separated, locally informed large-language-model (LLM) agent instances as a finite interacting stochastic system. Each persistent identity owns a categorical belief, a committed action, bounded private memory, a local field, an inbox and outbox, and typed local authority; a random-sequential scheduler offers update opportunities but does not choose the resulting state. We distinguish the complete augmented simulator state, its observable belief-action projection, and a rolling macroscopic representation. This separation exposes how omitted private memory can act as a hidden historical coordinate. We report collective order, uncertainty, dependence, an effective symmetric-layer energy, fluctuations, and coarse-grained path-reversal divergence. We first correct a derived-analysis error in an earlier frozen Qwen quench study without changing any trajectory. We then conduct a prospectively frozen two-family experiment with 48 trajectories, 34,560 local decisions, and six independent graph/environment clusters per model. A field reversal and restoration is paired with nominal evolution, genuine persistent memory, and a format-matched scrambled-history control. The independent-family quench effect is 42.263 (95% interval 24.316 to 59.303) within-model distance units. Genuine memory changes bias-adjusted reversal divergence relative to Markovized and scrambled-history controls by 0.05438 (0.03459 to 0.07548) and 0.04845 (0.02190 to 0.07526) nats per attempted update. The fixed early-to-late restoration contrast is 52.541 (35.406 to 68.673) distance units. All four effects meet their frozen criteria, although the memory contrasts are larger and more uniform in Granite than in Qwen. An out-of-sample kinetic surrogate identifies which response components admit a simple local closure and which remain model- and history-dependent. These finite-size results support a reduced statistical-mechanical description without interpreting decoding temperature as physical temperature, reference energy as literal energy, or projected path divergence as exact thermodynamic entropy production.

Keywords: statistical mechanics of agents; language-model agents; quench dynamics; entropy rate; multi-information; temporal asymmetry.

1 Introduction

Networks of language-model agent instances are interacting many-component systems with unusually complicated local transition rules. A component can retain private history, condition on a natural-language observation, receive only selected messages, and execute a

typed local action. Even when model weights are fixed, the collective trajectory depends on graph structure, asynchronous scheduling, message content, bounded memory, and decoding randomness. Population averages alone can hide this organization. The statistical-mechanics question is whether carefully defined reduced variables can connect local response to collective order, fluctuation, persistence, and driven reorganization.

The kinetic Ising model provides the canonical example in which stochastic local updates generate collective relaxation [1]. Related logit dynamics has an equilibrium stationary law when interactions derive from a potential [2]. Nonreciprocal interactions can instead produce collective structures and responses unavailable to reciprocal potential systems [3, 4]. State-space currents and trajectory reversal quantify time asymmetry in Markov processes [5–8]. Those exact constructions are references rather than assumptions here. An observable LLM belief–action process may fail first-order closure because a prompt can contain persistent private history and semantically structured packets. Coarse-graining can hide entropy production or generate estimator-dependent lower bounds [9–11]. We therefore use the more precise term *coarse-grained pathwise irreversibility* for our empirical block-reversal statistic.

Recent studies have begun to apply statistical physics to interacting LLM populations. Group size can change basins and collective misalignment [12]; topology and self/social weighting alter conformity [13]; intrinsic bias can dominate fitted neighbor coupling [14]; debate noise produces finite-size ordering crossovers [15]; and large populations can be summarized by a small family of fitted dynamical regimes [16]. Our contribution is complementary. We retain separate belief and committed-action layers, explicit local authority, inbox and outbox boundaries, random-sequential updates, persistent private memory, a Markovized comparison, and a controlled quench followed by a counter-quench. We place direct Qwen [17] and Granite [18] trajectories beside an out-of-sample kinetic closure.

The central claim is descriptive and explanatory, not performance-oriented. Statistical-mechanical variables provide a compact language for uncertainty, dependence, compatibility, fluctuation, and temporal response that is not contained in a single magnetization. No task reward, controller improvement, human response, or operational benefit is evaluated. Nor do two pinned models establish universality.

The sections below proceed from the system and its observables to the evidence hierarchy, direct results, reduced-description test, and interpretation.

2 State-separated LLM-agent networks and quench protocols

2.1 Augmented, local, and observed state

For agent i at attempted update t , the primary categorical coordinates are belief $b_i(t) \in \{-1, +1\}$ and committed action $a_i(t) \in \{-1, +1\}$. Its bounded local state also includes confidence, commitment status, workload, a memory code, at most three private history entries, an inbox, and an outbox. Display labels are counterbalanced and decoded back to latent spins. The microscopic projection is

$$Y_t = (\mathbf{b}_t, \mathbf{a}_t, \mathbf{c}_t, \mathbf{m}_t), \quad (1)$$

but this is not the complete simulator state.

Let

$$\Xi_t = (S_{1,t}, \dots, S_{N,t}, G_t, \mathcal{D}_t, q_t, R_t) \quad (2)$$

denote all agent-private states $S_{i,t}$, graph state G_t , delivery operator $\mathcal{D}_t$, quench phase q_t , and the explicitly specified randomness state R_t . With fixed model weights and tokenizer, reconstructed local prompts, and a defined scheduler, Ξ_t induces a time-inhomogeneous transition kernel under the external quench protocol. The evaluator’s map $Y_t = \phi(\Xi_t)$ deliberately omits some prompt history and randomness. Consequently Y_t and any rolling macrostate need not be Markov.

Figure 1 summarizes the information boundaries and the augmented-to-macroscopic map.

Augmented process Ξ_t and measured projections

$Y_t = \phi(\Xi_t); Z_t = \psi(Y_{t-w+1:t})$ (evaluator-side)

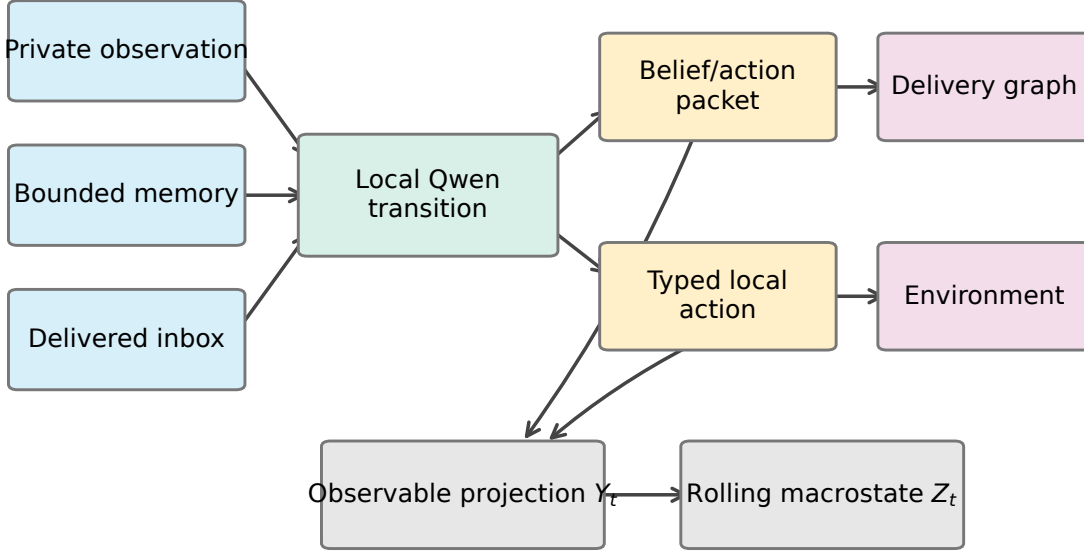


Figure 1: Agent information boundaries and the augmented-to-macroscopic map. The complete state Ξ_t includes private memory, graph and schedule state; the evaluator observes $Y_t = \phi(\Xi_t)$ and constructs rolling $Z_t = \psi(Y_{t-w+1:t})$. The scheduler offers a turn and delivers a packet but does not supply the model’s belief, action, memory, or signal.

2.2 Information boundaries and random-sequential update dynamics

At each attempted update, a random permutation gives every agent exactly one opportunity per sweep. Only the scheduled instance receives its private observation, current local state, currently delivered inbox, and, in a memory condition, its own bounded history. It returns a validated typed object containing belief, action, confidence, commitment, memory state, outgoing signal, and tool action. One bounded repair may correct syntax; an invalid result causes no centrally selected scientific action. A serialized packet is sent on one graph edge. Fingerprints of nonrecipient peer-private states are compared around every transition.

The corresponding authority and private-state checks are recorded in Appendix D.1.

2.3 Markovized, persistent, and scrambled-history conditions

The Markovized prompt is reconstructed from the explicitly recorded current state and does not carry free conversational history. The persistent prompt adds the instance’s last three own entries in the fixed format **time**, **belief**, **action**, **memory**. Thus two trajectories with identical visible $(\mathbf{b}_t, \mathbf{a}_t)$ can have distinct conditional future laws:

$$P(Y_{t+1} \mid Y_t, M_t) \neq P(Y_{t+1} \mid Y_t, M'_t). \quad (3)$$

If M_t is projected out, conditional dependence on older visible states can remain. This is a concrete mechanism for increased block reversal divergence, not a proof of physical dissipation.

The V15 scrambled-history control isolates prompt presence and approximate length. It uses only the same agent’s past opportunity times, while belief, action, and memory content are deterministically randomized from a frozen control seed. No donor agent, future time, or hidden environmental outcome is used. Genuine and scrambled entries have the same section, count, field names, and format. Token counts and their paired differences are measured; turn-by-turn equality is not assumed.

Prompt-balance diagnostics for the scrambled-history control are retained in Figure 12 of Appendix B.3.

2.4 Field quench, restoration, and matched controls

Each trajectory has 15 baseline sweeps, 15 intervention sweeps, and 15 restoration sweeps. In the field-quench arm the vector of balanced private fields changes sign at the first boundary and returns to its initial value at the second. Nominal trajectories retain the initial field. This follows the logic of a finite-system quench protocol, while making no quantum or thermodynamic-limit analogy. The initial change is the quench; restoration is the counter-quench.

Each family contributes six new graph/environment clusters. Each cluster has $N = 16$, a reciprocal modular fixed-degree graph, coupling $J = 0.8$, balanced disordered initialization, and four matched trajectories: nominal Markovized, field-quench Markovized, field-quench persistent memory, and field-quench scrambled history. Graphs, fields, initial states, label maps, update orders, recipient variates, and inference seeds are paired where technically meaningful.

3 Statistical-mechanical observables and inferential design

3.1 Order, overlap, effective compatibility, and fluctuations

Belief and action order and their overlap are

$$m_b = \frac{1}{N} \sum_i b_i, \quad m_a = \frac{1}{N} \sum_i a_i, \quad q_{ba} = \frac{1}{N} \sum_i b_i a_i. \quad (4)$$

Disagreement is the fraction of connected pairs with unequal states. For a symmetric comparison layer $W_s = (W + W^\top)/2$, we define

$$H_{\text{ref}} = -\frac{J_b}{2} \mathbf{b}^\top W_s \mathbf{b} - \frac{J_a}{2} \mathbf{a}^\top D_s \mathbf{a} - K \mathbf{a}^\top \mathbf{b} - \mathbf{h}^\top \mathbf{b} - \mathbf{g}^\top \mathbf{a}. \quad (5)$$

We report $e_{\text{ref}} = H_{\text{ref}}/N$. Equation (5) is an effective compatibility coordinate anchored in a symmetric reference model. We do not infer a Gibbs law for the LLM process and do not call this literal energy. Likewise $N \text{Var}(e_{\text{ref}})$ is an energy-fluctuation observable, not heat capacity unless an independently justified effective temperature exists.

The belief susceptibility is reported without physical-temperature normalization,

$$\chi_b = N \text{Var}(m_b), \quad (6)$$

within a prescribed rolling window.

3.2 Entropy, dependence, and the rolling macrostate

For discretized collective words x in a rolling window, Shannon configuration entropy is

$$S_{\text{config}} = - \sum_x \hat{p}(x) \log \hat{p}(x). \quad (7)$$

This is the usual plug-in Shannon functional [19]. The entropy rate is a bounded-history plug-in estimate of next-word unpredictability. Single-variable marginal entropy measures local occupancy; pairwise mutual information measures dependence between variables. Total correlation [20],

$$\mathcal{T} = \sum_{k=1}^{2N} H(X_k) - H(X_1, \dots, X_{2N}), \quad (8)$$

separates marginal uncertainty from synchronous joint dependence.

The plug-in joint entropy of 32 binary variables in a short window has a large finite-sample floor [21]. We therefore retain the raw estimate but independently circular-shift each variable 200 times. Shifts preserve each marginal occupancy and within-variable temporal ordering while breaking synchronous cross-variable relations. We report raw, null mean, their untruncated difference, and a normalized difference divided by the sum of marginal entropies. The same audit is applied to pairwise and edge mutual information. A negative adjusted estimate remains negative.

The primary five-sweep representation is

$$Z_t = (m_b, m_a, q_{ba}, e_{\text{ref}}, v_E, S_{\text{config}}, h_{\text{rate}}, \mathcal{T}_{\text{adj}}, I_{\text{pair,adj}}, I_{\text{edge,adj}}, \chi_b, \rho_b, D_b)_t. \quad (9)$$

Three- and seven-sweep windows are frozen sensitivities. This is an observable reduced representation, not a complete thermodynamic state.

3.3 Coarse-grained pathwise temporal asymmetry

For a discrete observed word sequence and block length L , let $\hat{P}_L(x_0, \dots, x_L)$ be the regularized empirical forward block law. We compute

$$\mathcal{I}_L = \frac{1}{L} D_{\text{KL}} \left[\hat{P}_L(x_0, \dots, x_L) \parallel \hat{P}_L(x_L, \dots, x_0) \right] \quad (10)$$

in nats per attempted update. The primary $L = 3$ estimator uses pseudocount 0.5. Values at $L \in \{2, 4\}$ and pseudocounts 0.1 and 1.0 are sensitivities. Five hundred time permutations provide a panel-specific finite-length floor, and the adjusted statistic is observed (10) minus

its mean floor. Matched arms share the same shuffle tape. The machine-readable sensitivity table retains both the empirical permutation-null interval and the Monte Carlo standard error of its mean. These quantify different objects: dispersion of the null statistic and numerical uncertainty in the estimated bias floor, respectively. The added uncertainty columns reproduce, rather than replace, the frozen floor mean and adjusted contrast.

For a fully observed stationary Markov chain with transition kernel P and stationary law π , the current

$$J_{xy} = \pi_x P_{xy} - \pi_y P_{yx} \quad (11)$$

can define an exact Markov entropy-production rate under the usual support conditions. We do not apply that interpretation to projected LLM trajectories. Persistent histories, semantic content, and bounded observations can violate first-order closure. Equation (10) is instead a coarse-grained time-reversal diagnostic and, under additional assumptions, may be a lower bound on hidden irreversibility [8, 10, 22].

3.4 Spatial correlation and finite-window persistence diagnostics

Three secondary observables resolve spatial organization, temporal persistence, and the shape of the finite-size order-parameter distribution. First, for a phase window W we define the connected belief correlation at graph distance d ,

$$C_b(d; W) = \frac{1}{|\mathcal{P}_d|} \sum_{(i,j) \in \mathcal{P}_d} [\langle b_i b_j \rangle_W - \langle b_i \rangle_W \langle b_j \rangle_W], \quad \mathcal{P}_d = \{(i, j) : i < j, d_G(i, j) = d\}. \quad (12)$$

Shortest paths use the actual reciprocal modular delivery support. The bracket averages post-update observations at attempted-update resolution within one phase of one trajectory; node pairs are averaged inside that trajectory and never counted as independent replicates.

Second, the normalized magnetization autocorrelation and its fixed truncation are

$$\begin{aligned} \rho_m(\ell) &= \frac{\langle [m_b(t) - \bar{m}_b] [m_b(t + \ell) - \bar{m}_b] \rangle}{\langle [m_b(t) - \bar{m}_b]^2 \rangle}, \\ \tau_{\text{int}}^{(\ell_{\text{max}})} &= \frac{1}{2} + \sum_{\ell=1}^{\ell_{\text{max}}} \rho_m(\ell). \end{aligned} \quad (13)$$

The primary descriptive truncation is $\ell_{\text{max}} = 2N = 32$ attempted updates; N and $3N$ are fixed sensitivities. Zero-variance phase series are left undefined rather than assigned artificial persistence. The quantity is a finite-window correlation sum reported in attempted-update units, not evidence of critical slowing down or an effective sample-size estimate. Its absolute short-lag scale includes persistence induced by changing at most one agent on each random-sequential update; matched arms share that update convention. Quench and restoration windows can also be nonstationary, so their sums are finite-window persistence diagnostics rather than equilibrium integrated correlation times.

Third, we calculate the Binder cumulant

$$U_4 = 1 - \frac{\langle m_b^4 \rangle}{3 \langle m_b^2 \rangle^2} \quad (14)$$

separately for each phase and trajectory [23]. Near-zero second moments are reported as undefined. With only $N = 16$, equation (14) is a finite-size distribution-shape diagnostic; no crossing, critical point, or thermodynamic-limit transition is inferred. A post-reconstruction

estimator sensitivity repeats the calculation over each full phase and its early and late halves. It compares the mean of the six trajectory-level cumulants with a pooled-moment construction; both uncertainty calculations resample complete graph–environment clusters, never updates.

3.5 Nominal geometry, recovery estimand, and exact cluster inference

For each held-out graph/environment cluster and model, the nominal manifold is fitted only from nominal trajectories in the other five clusters. Training medians impute nonfinite coordinates; training means and scales standardize features; shrinkage covariance with frozen ridge fraction defines a Mahalanobis-like distance. The nominal 95th-percentile threshold is computed from those training distances. The held-out cluster contributes neither a scale nor a threshold.

We summarize peak departure, integrated response, path length, final residual, and threshold re-entry. The prospective V15 recovery contrast is

$$\Delta d_{\text{rec}} = \bar{d}_{31:35} - \bar{d}_{41:45}, \quad (15)$$

which can be positive, zero, or negative. It replaces a historical maximum minus final-window quantity whose sign was structurally constrained.

A complete graph/environment trajectory cluster is the inferential unit.

The hypotheses are frozen as follows. H1 compares Granite field-quench peak departure with matched nominal evolution at one-sided $\alpha = 0.02$. H2 compares persistent with Markovized adjusted reversal divergence. H3 compares persistent with scrambled history. H4 tests the fixed recovery contrast. H2–H4 form a one-sided Holm family with total $\alpha = 0.03$. Exact cluster sign flips are the primary tests; deterministic 10,000 cluster bootstraps give intervals. The sign-flip enumeration is exact conditional on the observed paired magnitudes under a sign-symmetry null; it is not described as treatment- label randomization. Agent and time-point counts do not increase the inferential sample size.

The exact-test, multiplicity, and cluster-bootstrap details are retained in Appendix A.3.

4 Evidence hierarchy, correction, and frozen study roles

4.1 Memory discovery and prospective replication roles

Figure 2 keeps discovery, prospective replication, and the V15 cross-model extension in their distinct study roles.

4.2 Corrected V14 quench evidence and audit boundary

The V14 audit preserves every frozen prompt, completion, categorical choice, and trajectory. The original source computed a field panel’s recovery threshold from its own baseline, despite protocol text specifying a training-only threshold. Audit version 1.1 passes an explicit cluster-to- training-threshold map from leave-one-cluster-out fitting into recovery summaries. A versioned archive retains the committed historical JSON, tables, README, claims matrix, and macros.

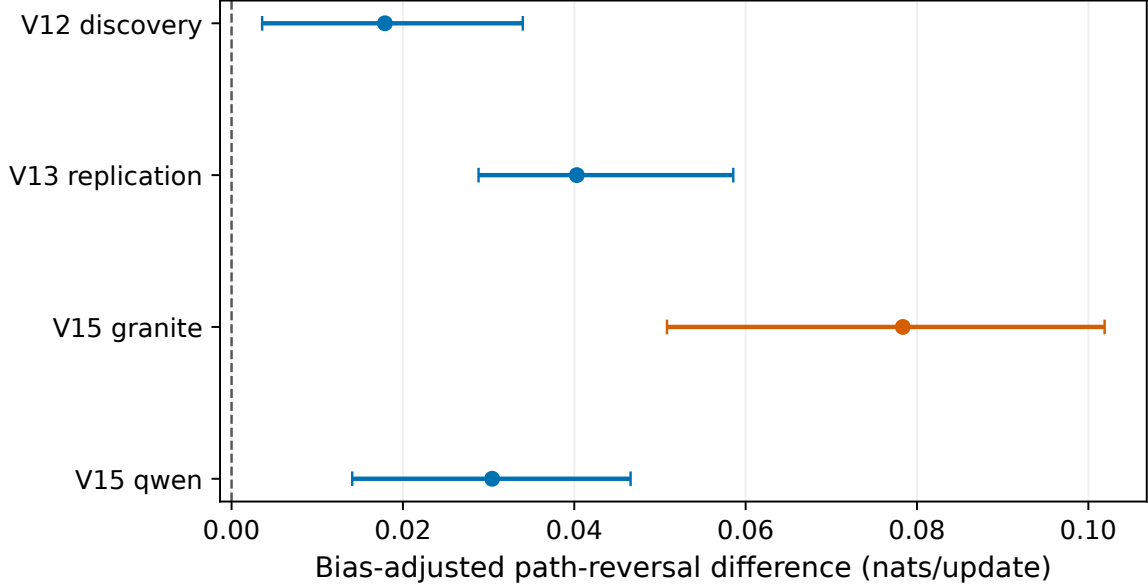


Figure 2: Discovery, prospective replication, and V15 cross-model memory contrasts are shown separately. Points are graph/environment-cluster effects; intervals resample complete clusters. V12 and V13 are not retroactively pooled into the V15 confirmatory interval.

The frozen H3 number was the maximum recovery distance minus the final-five mean. It remains numerically reproducible, but this construction is nonnegative for nearly every nonconstant trajectory. Its raw and Holm-adjusted sign-flip probabilities are retained as historical audit values and removed from inferential support. The corrected machine-readable fields distinguish whether the numerical criterion was met, whether a directional test is valid, and whether the time-resolved path is consistent with return.

Figure 3 displays the corrected V14 quench time series.

Cluster-level recovery and delayed-audit material is retained in Appendices A.1 and A.2, including Figures 9 and 10.

4.3 Frozen V15 cross-model confirmation and control logic

The original model is Qwen/Qwen2.5-7B-Instruct at immutable revision `a09a35458c702b33eeacc393d103063234e8bc28`. The independent family is IBM Granite-3.3-8B-Instruct at revision `51dd4bc2ade4059a6bd87649d68aa11e4fb2529b`. Both run with Transformers 4.55.4, PyTorch 2.8.0+cu128, bitsandbytes 0.47.0, NF4 double quantization, BF16 computation, decoding temperature 0.5, top- $p = 0.9$, and at most 96 generated tokens. The model’s decoding temperature is not an inferred thermodynamic temperature.

The engineering pilot is restricted to loading, schema validity, occupancy, both transition directions, timing, privacy, delivery, prompt-token balance, latency, and projected compute. It does not calculate a quench, memory, or irreversibility contrast. The formal design contains 48 trajectories and 34,560 attempted local decisions. Complete unfavorable trajectories are retained.

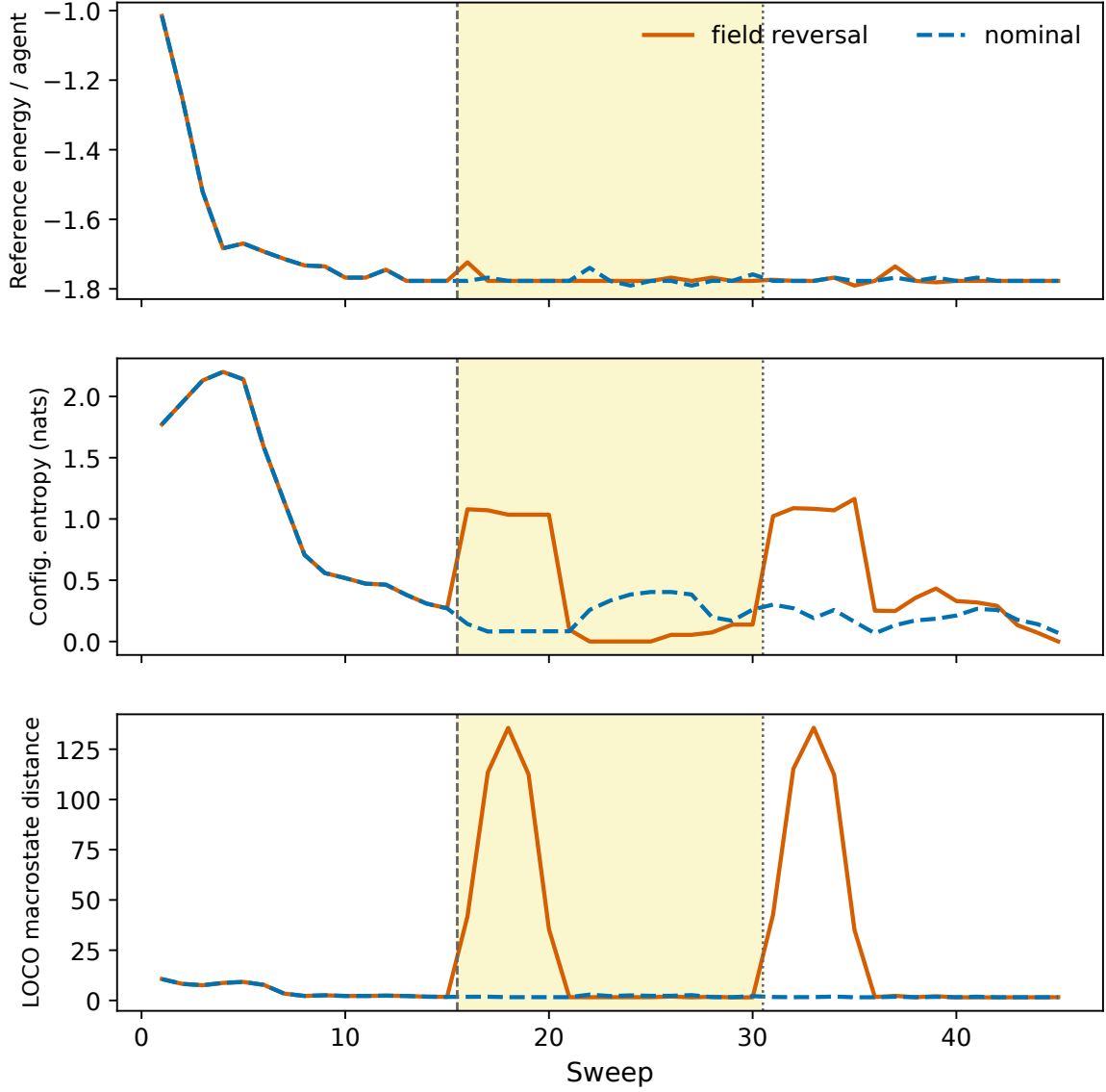


Figure 3: Corrected V14 time series for nominal and field-reversal trajectories. The yellow interval marks the quench; the dotted boundary marks restoration. Reference energy is effective, entropy is configuration diversity, and distance is measured in the cluster-excluded nominal geometry.

5 Quench response, restoration, and finite-size organization

5.1 Cross-model field response

The Granite-family H1 contrast is 42.263 distance units (95% interval 24.316 to 59.303); its frozen disposition is supported. Figure 4 displays model-specific distances rather than pooling their absolute scales. A positive contrast indicates a field-driven departure beyond nominal variation; a nonpositive or uncertain contrast would delimit cross-family replication. The complete cluster table is retained regardless of direction.

All six Granite cluster effects are positive. The separately retained Qwen contrast is also positive in all six clusters, with a mean field-minus-nominal peak of 112.691 units. These absolute values are not compared across models: each leave-one-cluster-out geometry is standardized and fitted within model.

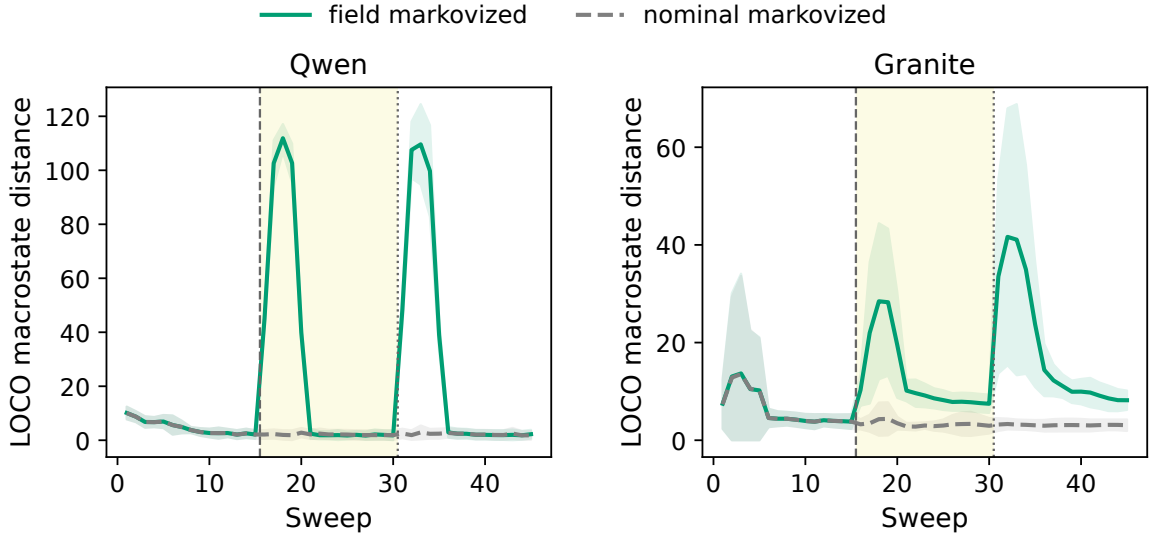


Figure 4: Model-specific field-quench and nominal trajectories. Curves are means over six graph/environment clusters and bands show between-cluster dispersion. Nominal fits, scaling, covariance, and thresholds exclude the displayed cluster.

5.2 Restoration and threshold re-entry

Restoration H4 is 52.541 distance units (95% interval 35.406 to 68.673), with frozen disposition supported. Unlike the historical V14 H3 statistic, this fixed-window quantity permits either sign. Recovery time and threshold crossing remain trajectory descriptors, not an equilibrium relaxation theorem. The quench and counter-quench paths need not retrace one another because messages, actions, and history alter the state reached at restoration.

The fixed-window decline occurs in every model-cluster pair, but it does not imply complete return within the observation horizon. Every Qwen trajectory re-enters its training-nominal threshold six sweeps after restoration. Only one of six Granite trajectories re-enters by sweep 45; the other five retain final-five means above their own thresholds. Thus H4 supports motion toward the restored nominal regime, while Granite exhibits slower or incomplete finite-horizon recovery.

5.3 Spatial reorganization under the quench

Equations (12)–(14) are secondary descriptive analyses of the same frozen trajectories; they neither modify the confirmatory estimands nor introduce additional simulation arms. They were specified during reconstruction after the original V15 aggregate results were known and are therefore not a new prospective hypothesis family. Connected correlation matrices are first calculated within every complete model–graph–environment trajectory and phase. Their graph-distance profiles are then summarized over the six independent clusters. Figure 5 shows cluster-averaged matrices for the persistent-history quench window and phase-resolved distance profiles. Community boundaries are fixed by the graph construction rather than selected from the observed correlations.

At graph distance one, the persistent-history quench-minus-baseline connected correlation changes are 0.00238 (−0.01750 to 0.02277) for Qwen and −0.02032 (−0.05486 to 0.00947) for Granite. These cluster-level descriptive intervals quantify finite-window reorganization; their signs are not assigned a confirmatory disposition.

5.4 Finite-window persistence and order-parameter shape

Figure 14 in Appendix C combines the full autocorrelation curve with the fixed-lag integral in equation (13), the Binder statistic in equation (14), and empirical magnetization occupancy. This combination prevents a Binder value from being interpreted without its underlying finite-window distribution. It also separates persistence of the projected collective coordinate from the block-reversal statistic: the former is time-symmetric two-point dependence, whereas the latter compares forward and reversed path words. The accompanying source table also retains full-, early-half-, and late-half Binder estimates under cluster-first and pooled-moment rules. Their differences diagnose finite-window and aggregation sensitivity rather than selecting whichever construction yields the larger separation.

At the fixed two-sweep truncation, persistent-minus-Markovized recovery τ_{int} changes are −2.46995 (−5.79214 to 0.17724) updates for Qwen and −4.14109 (−19.88286 to 11.60067) updates for Granite. The field-Markovized quench-minus-baseline Binder changes are −4.80504 (−7.42824 to −2.20832) and 0.16643 (−0.09447 to 0.52063), respectively. These estimates are reported whether positive, negative, or interval-compatible with zero.

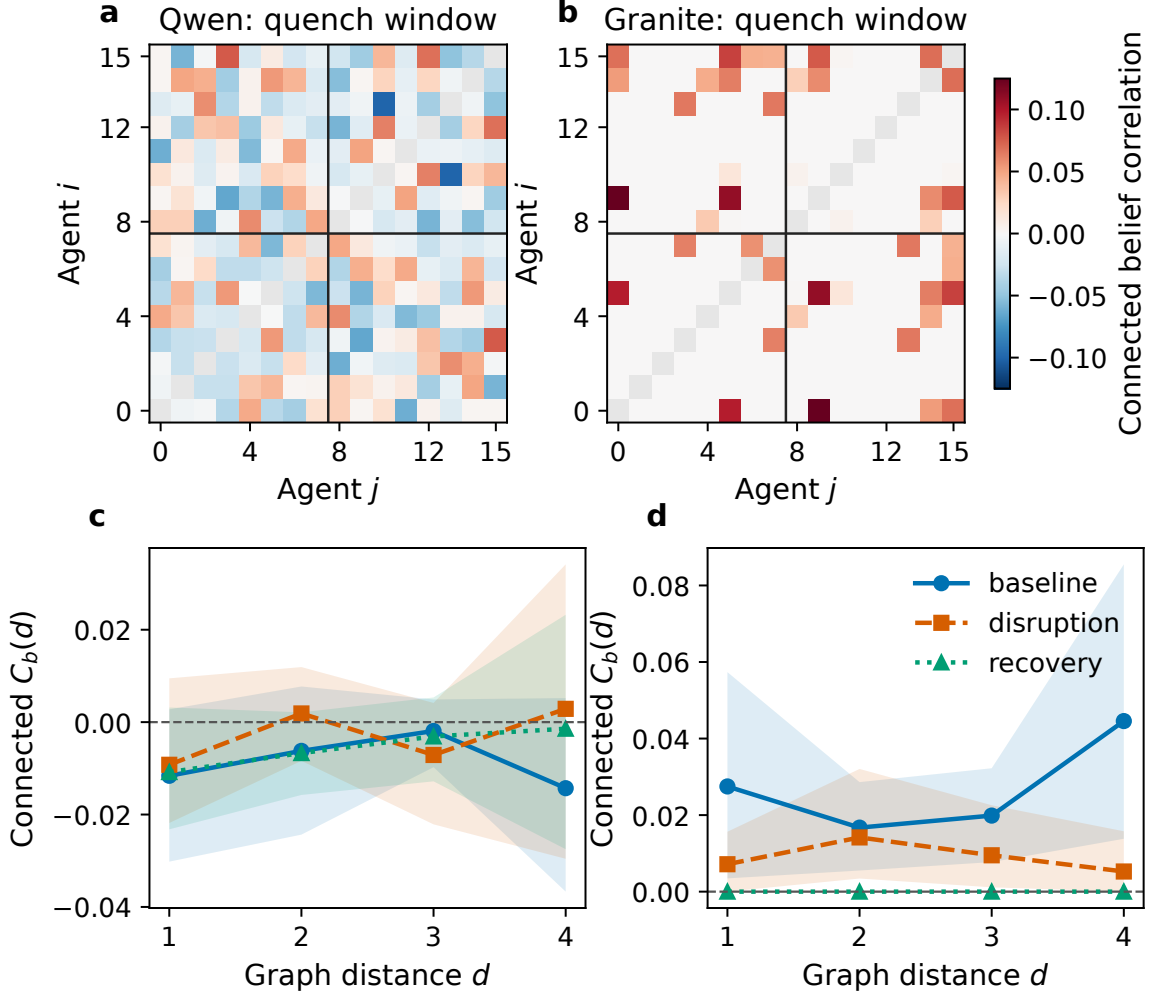


Figure 5: Connected belief correlations in the persistent-history arm. The upper panels average complete 16×16 connected-correlation matrices over all six clusters in each model; black rules mark the two predefined communities. Agent indices align across graph realizations only through those community memberships, so fine node-level texture in an averaged matrix has no cross-realization identity interpretation. The lower panels show equation (12) by graph distance, with 95% complete-cluster bootstrap intervals. Individual pairs and updates do not enter the inferential sample size.

6 Memory and projected temporal asymmetry

6.1 Persistent versus Markovized history

Relative to Markovized state, genuine memory changes adjusted block divergence by 0.05438 nats per attempted update (95% interval 0.03459 to 0.07548); the frozen disposition is supported.

H2 asks whether history matters relative to an explicit current-state prompt.

6.2 Genuine versus scrambled-history control

Relative to the scrambled-history placebo, the change is 0.04845 (0.02190 to 0.07526), disposition supported.

H3 is more stringent: it asks whether the temporally related content matters beyond adding a similarly shaped prompt section.

Several arm-level adjusted values are negative because their observed block divergence lies below the corresponding finite-sample shuffle-floor mean. H2 and H3 are paired differences between these untruncated adjusted statistics; their positive contrasts do not establish positive absolute entropy production in either arm.

No exact thermodynamic interpretation follows from a positive result. The observed block words omit natural-language content and some private state. Conversely, a null H3 would be informative: prompt presence, bounded memory depth, model-specific history use, or estimator power could explain the V12 and V13 contrast without a reproducible content-specific effect. Block-length, pseudocount, conditional-memory-depth, occupancy, and token-balance tables support this distinction.

Figure 6 displays the paired cluster contrasts for the two pinned model families.

The prompt-balance diagnostic in Figure 12 records the scope of the format and approximate-length control.

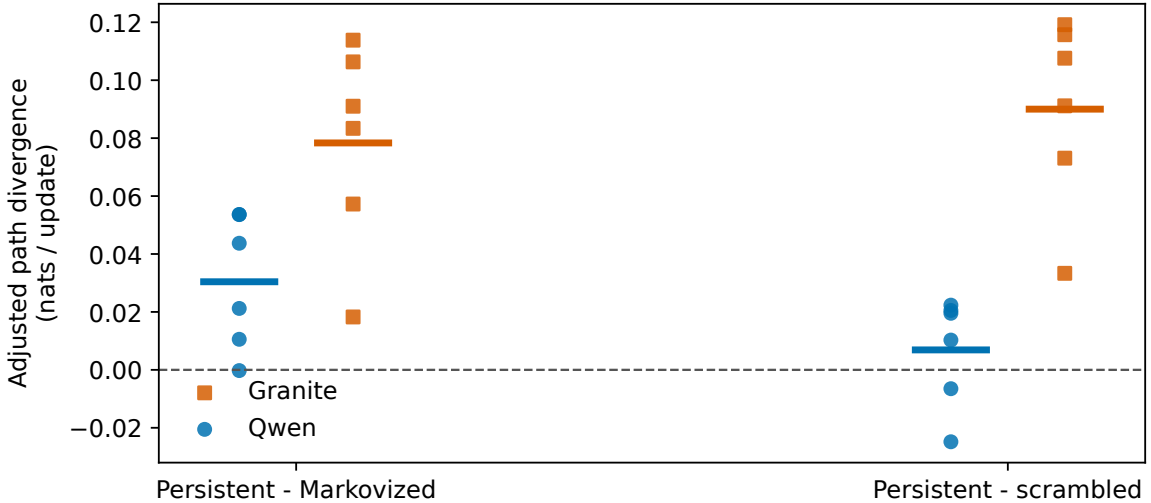


Figure 6: Per-cluster genuine-memory contrasts for both pinned model families. Short horizontal segments indicate model means. The scrambled-history arm controls prompt section and approximate length while destroying temporal relation to the current trajectory.

6.3 Model heterogeneity and estimator dependence

The prospectively pooled contrast retains substantial model heterogeneity. For persistent minus Markovized state, model-specific means are 0.03041 for Qwen (five of six cluster effects positive) and 0.07835 for Granite (six of six positive). For persistent minus scrambled history they are 0.00689 for Qwen (four of six positive) and 0.09001 for Granite (six of six positive). The model-specific Qwen scrambled-control sign-flip value is 0.21875 and is not separately confirmatory; the frozen pooled H3 remains supported because model-stratified cluster pairs were the prespecified units. This decomposition supports cross-family direction for the Markovized contrast but not a claim that content-specific memory effects are uniform across model families.

Estimator sensitivity further limits the scale claim. The pooled memory contrasts remain positive over most prespecified block-length and pseudocount combinations, but their magnitude varies substantially; at block length four and pseudocount one, persistent-minus-scrambled is only 0.00024 nats per update and the Granite component is slightly negative. Conditional-memory information is likewise not monotonically larger in the persistent arm. Accordingly, the robust statement concerns the frozen coarse graining, its bias floor, and paired controls rather than an estimator-independent entropy- production rate.

The block-length, pseudocount, and shuffled-floor sensitivities are displayed in Figure 11 of Appendix B.2.

6.4 Confirmatory synthesis across H1–H4

Figure 7 places the four frozen effects and their cluster-bootstrap intervals in a single confirmatory synthesis.

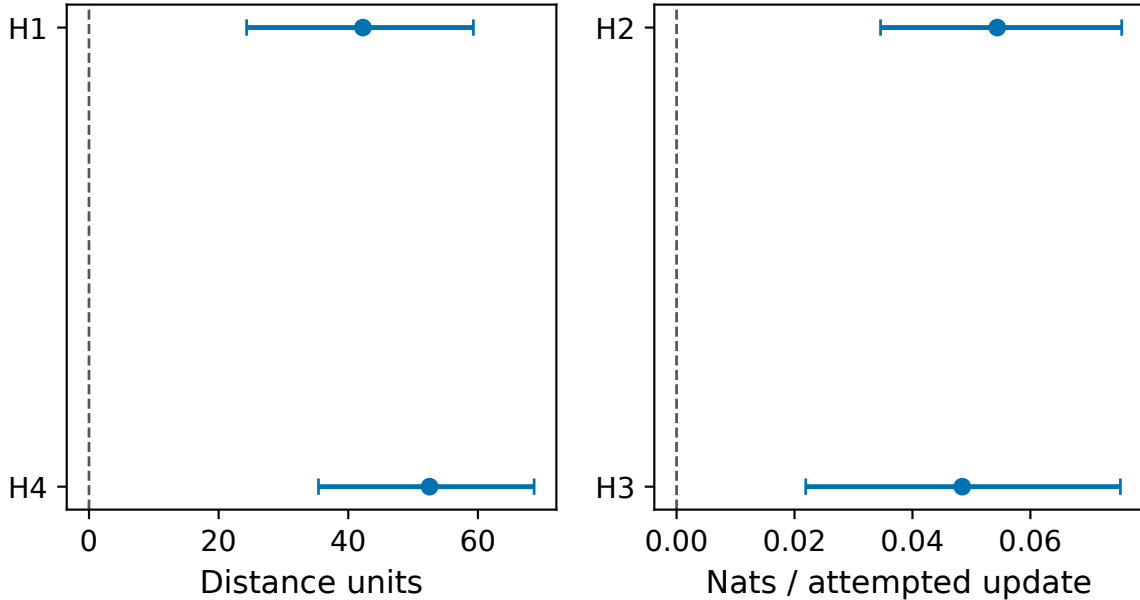


Figure 7: V15 frozen effects and 95% cluster-bootstrap intervals. Distance and information units occupy separate panels; no claims forest places incompatible quantities on one numerical axis. Dispositions use exact cluster sign flips and the frozen multiplicity allocation.

7 Reduced dynamical descriptions and closure limits

7.1 Information retained beyond order

Magnetization records directional order but can remain near zero when half the population reorganizes in opposing communities. Configuration entropy then records collective state diversity; marginal entropy records local occupancy; total correlation distinguishes independent uncertainty from coordinated dependence; effective energy records compatibility with the symmetric reference; susceptibility and energy variance record fluctuations; correlation time records persistence; and path reversal records temporal ordering. These quantities are complementary only when their estimators and scales are kept distinct.

The V14 representation experiment uses transparent multinomial logistic regression with leave-one-cluster-out preprocessing. The delayed permutation audit tests both absolute balanced accuracy and the increment over order-only features. This is not a claim that entropy outperforms all ordinary summaries. It tests whether a predefined collective representation retains condition information discarded by a reduced representation. The field-quench trajectory supplies a mechanistic example: energy, dependence, entropy rate, and fluctuation can change while mean magnetization remains balanced.

7.2 Out-of-sample kinetic surrogate

We fit logistic belief and action responses only to the immutable V13 microscopic-response table. The belief model includes private field, delivered neighbor field times controlled coupling, previous belief, and previous action; the action model includes updated belief and previous action. No coefficient is refitted to a V14 or V15 quench trajectory. The surrogate is run on the same graph generator, initial-state generator, asynchronous update tape, field schedule, and intervention times. It is a deliberately low-dimensional kinetic-Ising comparison rather than a fitted description of every asymmetric or history-dependent response [24].

Figure 8 compares the direct V14 trajectories with the out-of-sample kinetic surrogate.

In the five-coordinate comparison geometry, the direct field-minus-nominal peak-response difference is only 0.142 units on average and is nonzero in one of six clusters, whereas the surrogate difference is 1.467 units and is positive in four clusters. The corresponding field-minus-nominal energy-entropy route-area differences are 0.043 and 0.695. The direct shared response peaks at the quench boundary in one cluster and is otherwise dominated by initial relaxation; the surrogate places its peak at the quench or counter-quench boundary in four clusters. Both descriptions return close to their own final shared-response baseline, but the surrogate neither reproduces the direct amplitude nor its timing. This also explains why the richer V14 macrostate distance can show a large quench pulse that a short kinetic closure over a small shared coordinate set misses.

7.3 Captured response, failure modes, and the size-context boundary

This comparison identifies three possible failure modes. First, a homogeneous logistic neighbor coefficient cannot capture state-dependent semantic use. Second, action and belief persistence can introduce lags not represented by a single synchronous mean field. Third, hidden memory and workload can produce different future laws at the same projected spin

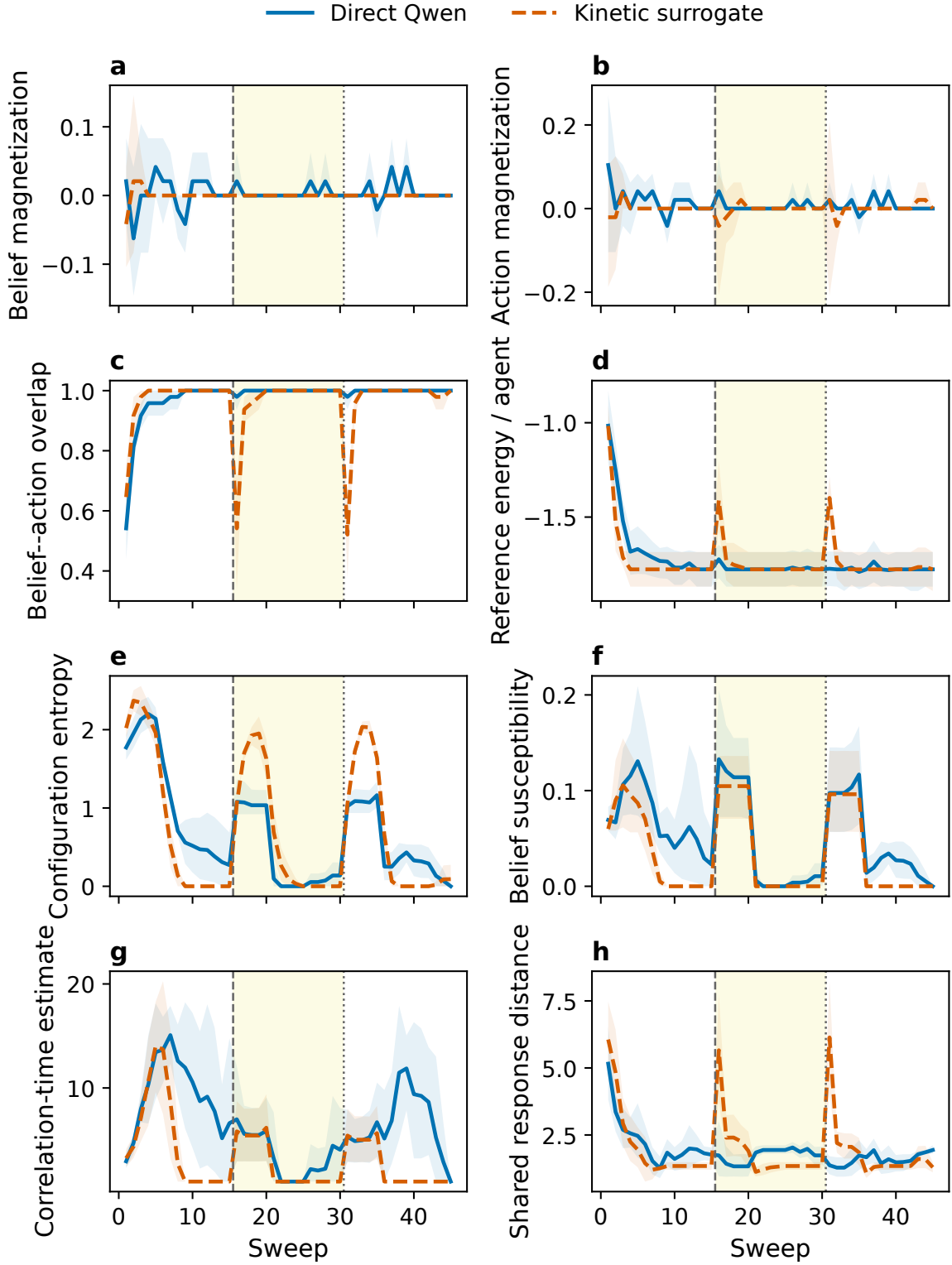


Figure 8: Out-of-sample V14 field-quench comparison between direct Qwen trajectories and the V13-fitted kinetic surrogate. Matching direction, peak timing, normalized amplitude, recovery, and route area are assessed separately. Surrogate failure is evidence about the limits of the closure, not a failed LLM experiment.

configuration. Adding unconstrained parameters after observing quench trajectories would obscure these limits, so the closure remains low-dimensional and out of sample.

CPU trajectories at $N \in \{8, 16, 32, 64\}$ supply a dense effective-model size context. They are not direct LLM finite-size scaling and cannot establish a critical exponent or thermodynamic-limit phase transition. The direct $N = 16$ paths remain empirical anchors.

The effective-model size context is displayed in Figure 13 of Appendix B.4.

8 Discussion

8.1 Literal stochastic process versus effective statistical mechanics

The formulation has a literal stochastic-process component and an effective statistical-mechanics component. Agent instances, local prompts, packet delivery, update schedules, and categorical state changes are implemented objects. Magnetization, Shannon entropy, mutual information, total correlation, and path KL are mathematical observables of recorded data. Reference energy is effective because the LLM transition rule is not derived from its Hamiltonian. Decoding temperature is a model-generation setting. No free-energy-like coordinate is emphasized because an independently identified effective temperature is unavailable.

8.2 Projected memory as hidden history

The distinction between complete augmented state, observable projection, and rolling macrostate explains why private memory can generate projected history dependence.

8.3 Finite-size, model, topology, and estimator limitations

Several limitations bound the claim. First, two 7B instruction-model families do not establish universality. Second, $N = 16$ direct systems are finite; no thermodynamic-limit transition is claimed. Third, the binary projection compresses semantic messages and private state, so temporal asymmetry is a coarse-grained diagnostic. Fourth, only a reciprocal modular topology and one coupling/noise anchor are used in V15. Fifth, a scrambled-history placebo matches format and approximate token distribution but cannot make semantic content exchangeable. Sixth, short-window entropy and dependence estimates remain coarse-graining dependent even after a null-floor audit.

8.4 Negative results, scope, and future work

The study also preserves negative boundaries. V12 found no reliable increase in irreversibility from degree- and traffic-matched nonreciprocity. V14 partition and message-corruption trajectories remained close to nominal under their tested construction. A simple kinetic closure missed parts of the direct coupling response. These limits prevent the broad assertion that any directed graph, entropy statistic, or Ising analogy is automatically useful.

Further work requires more model families, direct sizes and topologies, longer stationary trajectories, and observable-state augmentation before thermodynamic-limit or exact stochastic-thermodynamic claims would be warranted.

9 Conclusions

State-separated LLM-agent instances form a measurable interacting stochastic system whose collective trajectories admit a disciplined reduced description. Controlled field quench and restoration create a common test of order, uncertainty, dependence, effective compatibility, fluctuation, and temporal response. V15 adds an independent model family and a format-matched history control to the V12–V14 evidence. The pooled prospective memory effects are positive, while the weaker Qwen scrambled-control decomposition, Granite’s incomplete threshold re-entry, and estimator sensitivity remain explicit boundary results.

The statistical-mechanical contribution is therefore not that entropy makes agents perform better. It is a formulation and measurement framework that connects local language-conditioned transitions to collective finite-size dynamics, exposes what population averages discard, and makes failures of simple closures explicit.

Reproducibility and data availability

The repository records immutable model revisions, package versions, prompt and schema hashes, graph and seed manifests, control-tape hashes, frozen protocol, analysis configuration, aggregate tables, figure source data, vector PDFs, and content-addressed manifests. Raw prompts, completions, and full trajectories remain outside Git because they are large and potentially sensitive; their tree hashes and replay results are repository-facing. Every trajectory can be regenerated through content-addressed model decisions. The original sealed V15 execution used 19.19 measured generation GPU-hours; the fresh from-scratch reconstruction accounted for at least 49.41 measured generation GPU-hours under the same scientific protocol. This runtime difference is an environment-dependent reproducibility cost rather than a scientific outcome. The original external raw tree did not survive a later Pod replacement. The fresh call records replay internally at decision resolution and their frozen aggregate tables, calls, and token counts are compared with the committed reference package; historical call-file digest identity cannot be checked without the deleted original raw tree. The content-addressed accounting audit separately identified 197 decision records (0.328 generation GPU-hours) from any interrupted atomic panels; these records contribute to compute accounting but not to a scientific trajectory. One additional model generation failed after inference but before its atomic raw record could be written. It is counted as 1 infrastructure call, but its tokens and latency are unavailable; fresh reconstruction token, GPU-hour, and cost totals are therefore measured lower bounds. No chain of thought was requested or retained.

The manuscript uses standard 12-point LaTeX, which current IOP guidance accepts alongside the optional `iopjournal` class. Figures are embedded vector PDFs with source CSVs and are rendered at 300 DPI for inspection.

AI-assisted preparation declaration

AI-assisted software and text preparation was used under human-directed protocol constraints. All scientific estimands, code paths, numerical tables, citations, figures, and claims remain subject to author verification. No language-model output is treated as human evidence.

A Study chronology, V14 correction, and frozen inferential details

A.1 Training-only threshold correction and historical H3 invalidation

Figure 9 retains the cluster-level recovery audit under training-only thresholds.

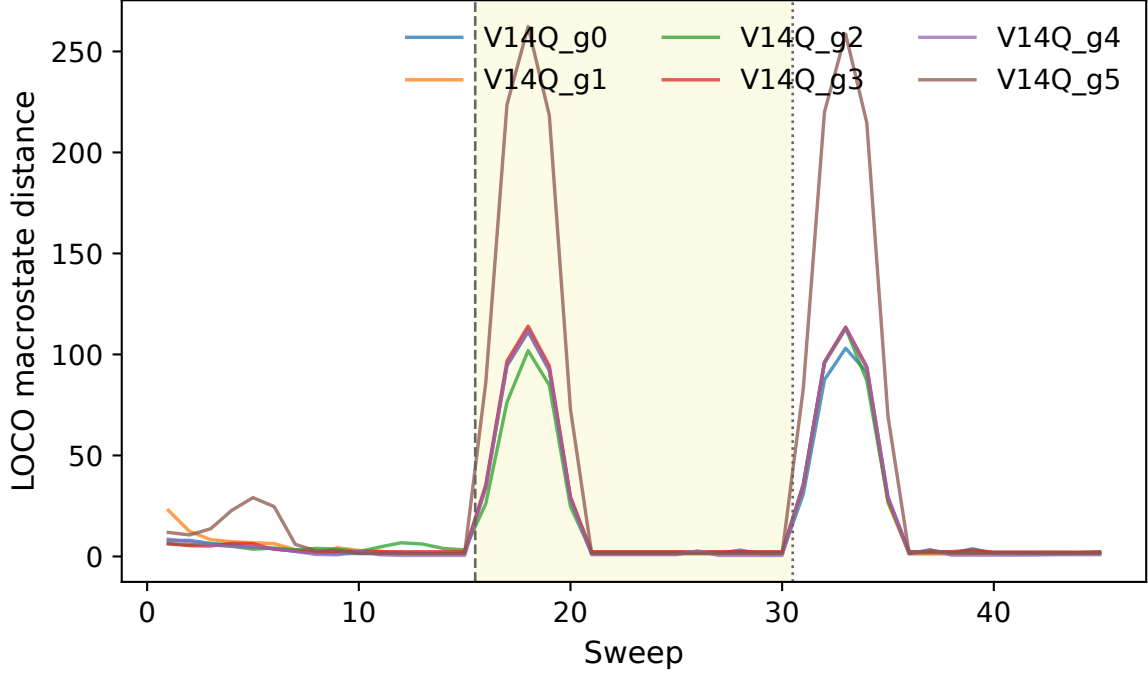


Figure 9: All V14 field-reversal cluster trajectories, including the high-response cluster. The derived audit replaces the held-out-panel threshold with the training-cluster nominal threshold. The historical H3 sign statistic is not used as directional evidence.

A.2 Delayed V14 permutation and observable-deletion audit

The audit also executes the omitted 10,000 cluster-preserving label permutations. All four panels stay together within each cluster, and every replicate refits imputation, scaling, logistic regression, and leave-one-cluster-out predictions. Three-, five-, and seven-sweep nominal geometries are fitted independently. Each coordinate and observable family is deleted in turn. Dependence estimates receive circular-shift null floors. These are delayed prespecified sensitivities, not a prospective V14 experiment.

The V14 representation permutation retains the four condition panels in every cluster and permutes labels only within cluster. For each of 10,000 replicates, every leave-one-cluster-out fold refits imputation, scaling, and the multinomial logistic model. Test-cluster features do not enter preprocessing.

Figure 10 records the delayed dependence and full-pipeline permutation audit.

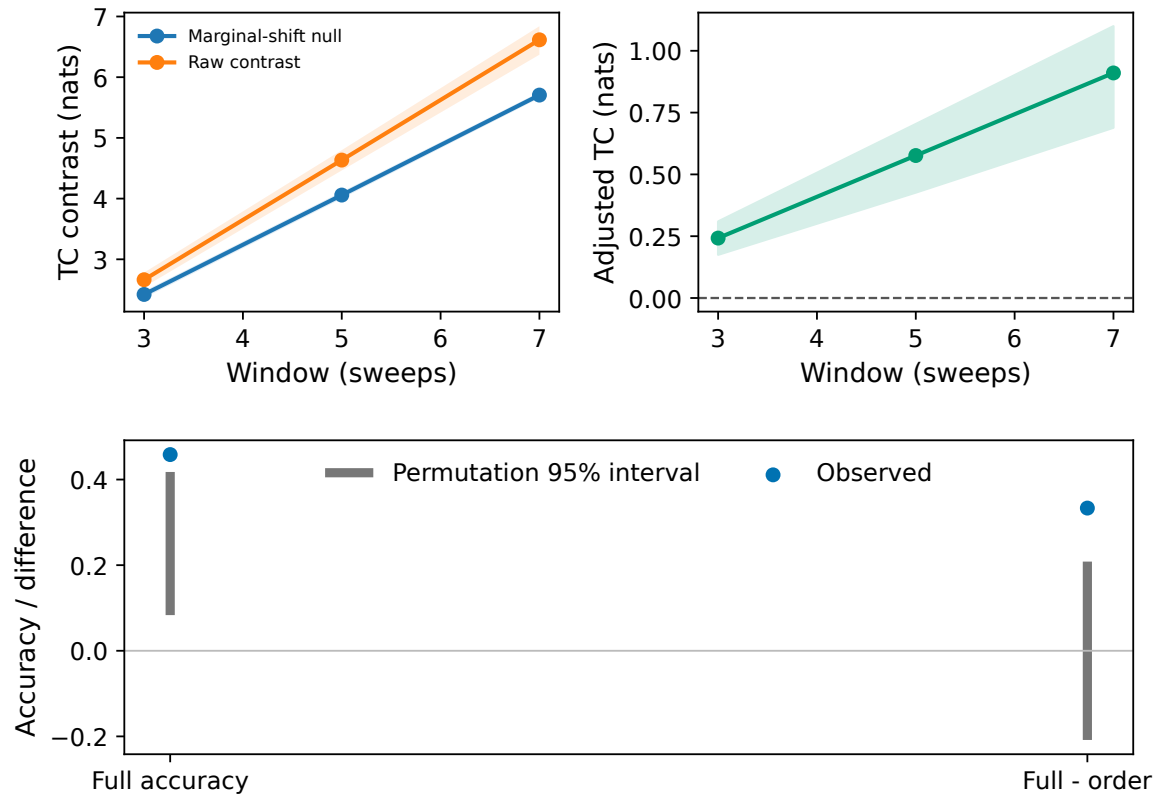


Figure 10: Delayed prespecified dependence and permutation audits. Raw and bias-adjusted dependence, window sensitivity, and cluster-preserving nulls are shown explicitly. These audits were completed after formal outcomes because of implementation omissions.

A.3 Frozen H1–H4 tests, multiplicity, and cluster bootstrap

For $n \leq 20$ paired clusters, the one-sided sign-flip probability enumerates all 2^n sign assignments of the observed cluster effects. The cluster bootstrap samples complete graph/environment clusters with replacement and never resamples time points as independent observations. H1 receives alpha 0.02; H2–H4 receive a Holm-adjusted family alpha 0.03. The allocation was fixed before formal outcomes. Intervals are uncertainty summaries and do not replace the exact paired tests.

B Estimator, control, and closure sensitivities

B.1 Rolling dependence and marginal-shift null construction

Total-correlation nulls independently circular-shift each coordinate by a nonzero offset where possible. This retains marginal occupancy and each coordinate’s internal temporal ordering while breaking synchronous dependence. The adjusted estimate is not truncated. Alternative rolling windows rebuild the words, covariance, training threshold, and held-out distance rather than subsampling a five-sweep output.

B.2 Block-reversal estimator and floor sensitivity

For block reversal, time-shuffled floors quantify finite-length occupancy bias. They do not prove stationarity. Results at block lengths two, three, and four expose finite-history sensitivity; conditional mutual information at history depths one through three provides a separate diagnostic of incomplete first-order closure.

Figure 11 displays the frozen estimator-sensitivity grid.

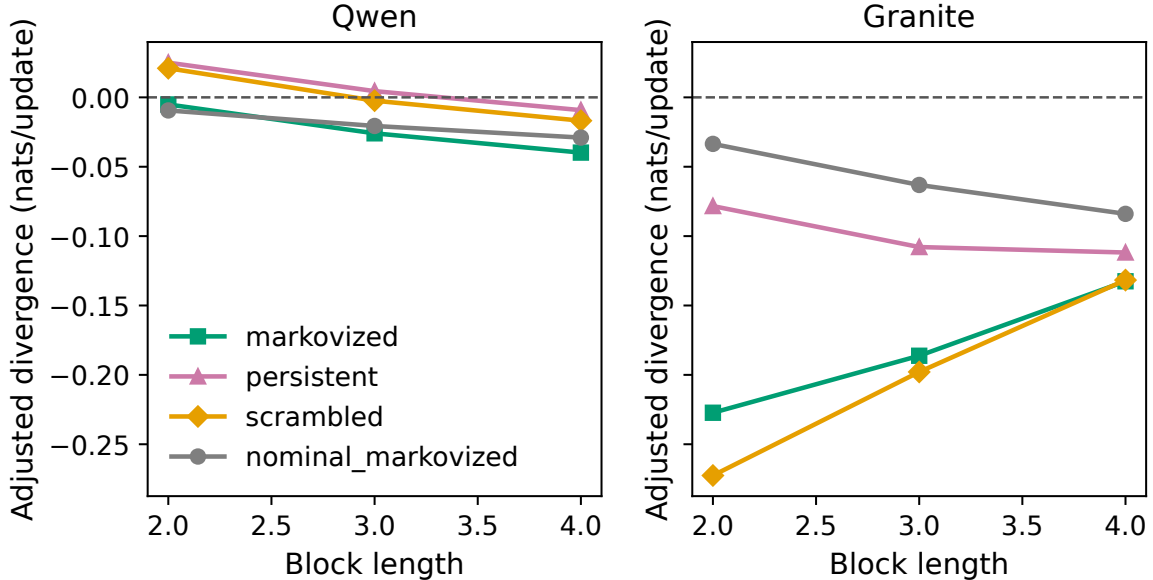


Figure 11: Block-length, pseudocount, and shuffled-bias-floor sensitivity for bias-adjusted block reversal divergence. Memory contrasts can be checked against estimator choices; observable coarse-graining and finite length remain limiting.

B.3 Scrambled-history prompt-balance control

Figure 12 records the frozen prompt-balance diagnostic.

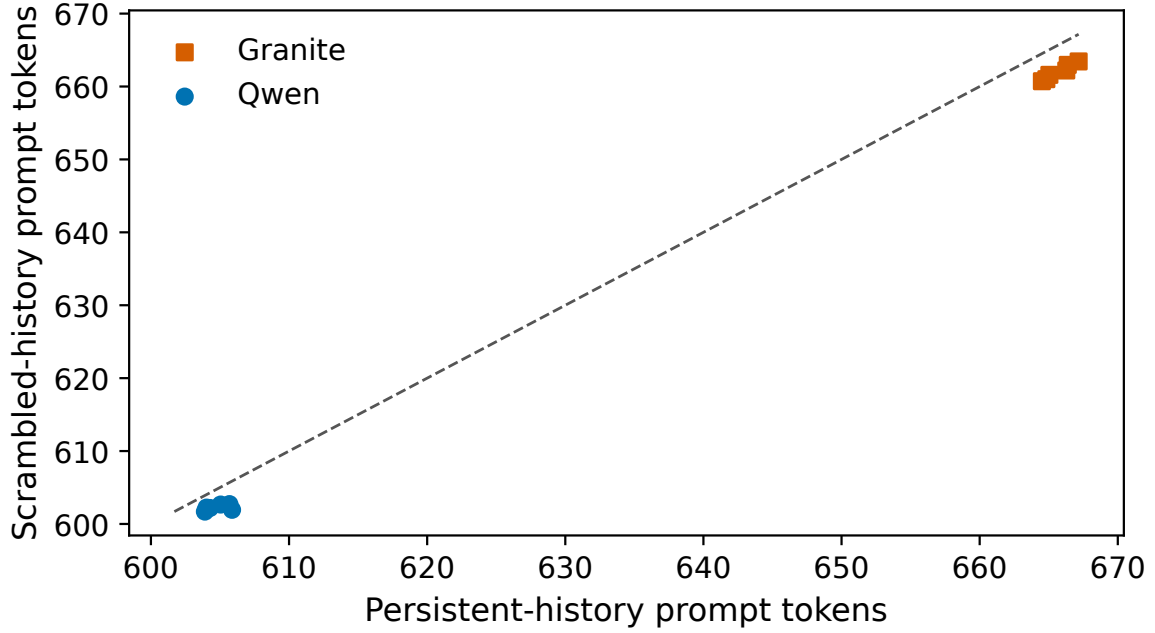


Figure 12: Scrambled-history prompt-length control. Per-cluster mean token counts verify that prompt length and format are approximately matched; semantic content cannot be exactly token-matched turn by turn.

B.4 Reduced-model size context

Figure 13 places the direct-system anchor beside the frozen effective-model size context.

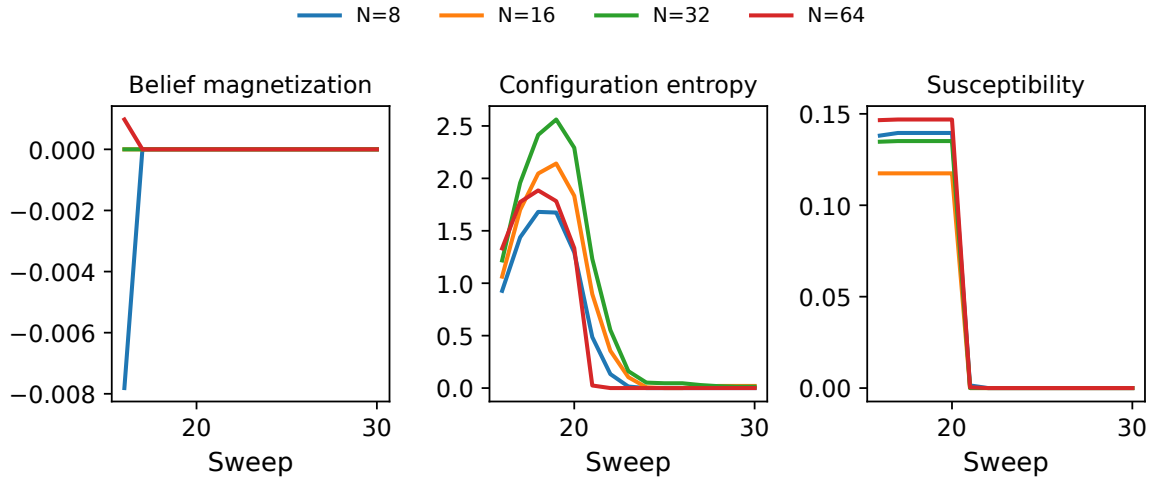


Figure 13: Direct $N = 16$ anchors in a denser inexpensive effective-model size context. These are CPU kinetic-surrogate quench responses, not direct-LLM finite-size scaling.

C Finite-window persistence and Binder diagnostics

C.1 Autocorrelation construction and truncation

The diagnostics below accompany equations (13) and (14) of the main text. Each quantity is computed within one complete model–graph–environment trajectory before uncertainty is summarized across the six clusters per model. The autocorrelation sum uses the fixed two-sweep truncation, while one- and three-sweep truncations remain machine-readable sensitivity analyses.

C.2 Binder trajectory-first and pooled-moment sensitivity

The Binder statistic describes the shape of a finite $N = 16$ magnetization distribution. The source-data table additionally retains full-phase versus early/late half-phase estimates and cluster-mean versus pooled-moment rules. It is not a crossing analysis or evidence of a phase transition.

C.3 Persistence, Binder, and occupancy display

Figure 14 displays the accompanying persistence, Binder, and occupancy diagnostics.

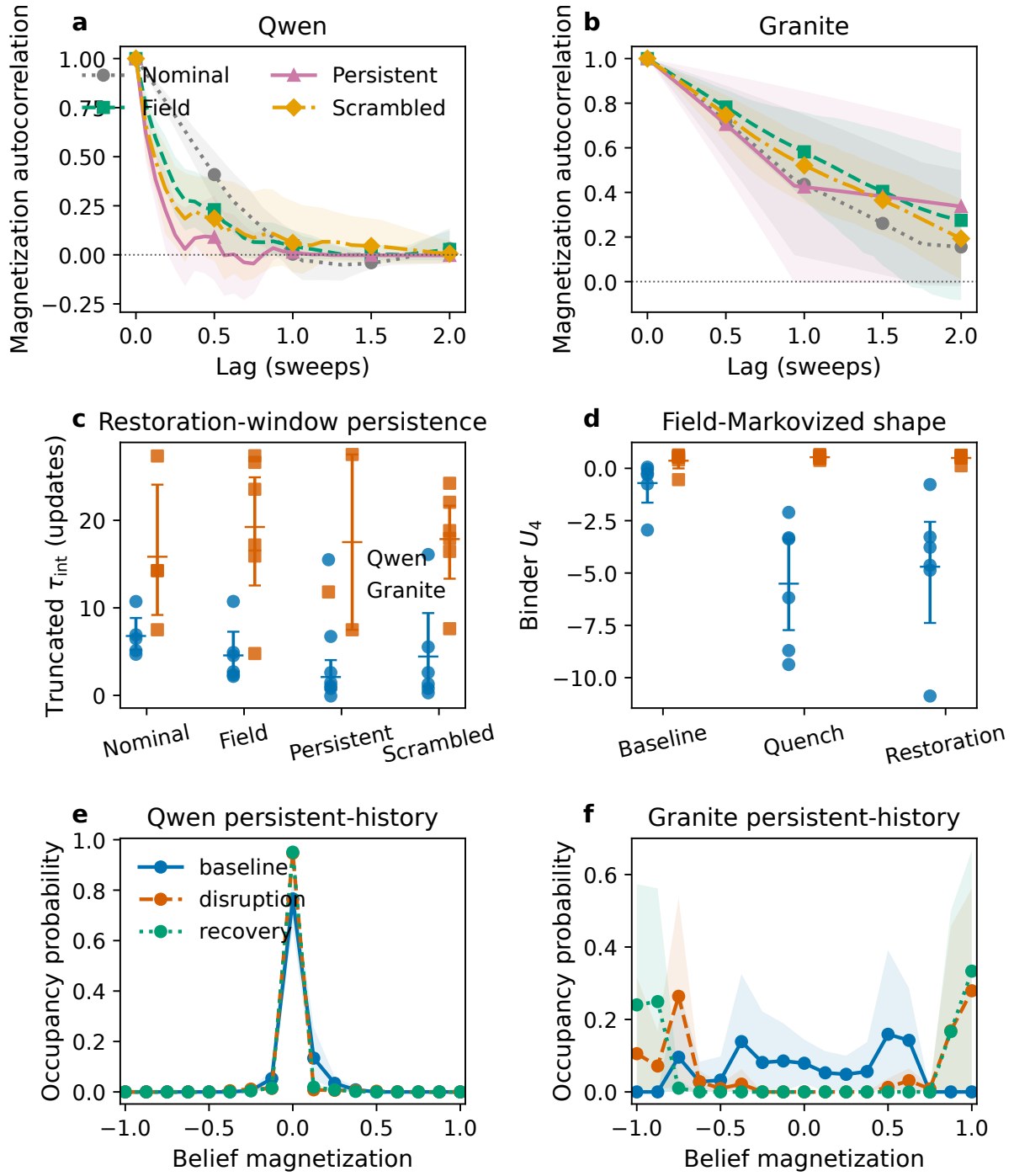


Figure 14: Finite-window persistence and order-parameter shape. Autocorrelation curves and their 95% intervals compare the nominal late window with the three field arms using complete clusters; integrated correlation sums show all cluster values at the fixed two-sweep truncation. Binder cumulants are paired with empirical belief-magnetization occupancies so their finite-sample meaning remains visible. These diagnostics do not establish critical slowing down or a phase transition.

D Implementation integrity, authority, privacy, and provenance

D.1 Agent authority and information-boundary checks

An agent’s response is causally consequential: it changes the scheduled instance’s recorded belief, action, commitment, memory code, signal, and typed tool consequence. The scheduler validates but does not replace those fields. Every transition records the scheduled identity, recipient, packet bytes, prompt hash, output-record hash, and fingerprints of unrelated peer-private state. Tests perturb one instance’s memory or decision provider while confirming that peer-private stores remain unchanged.

D.2 Arm independence, replay, and artifact integrity

Every experimental arm instantiates a fresh agent-network state from its frozen panel seeds. Model weights and the tokenizer are shared read-only for throughput, but conversational history, generation key/value caches, mutable agent objects, and scientific random-number state are not carried between arms. Each generation resets the frozen per-decision seed; record counters and provider accounting do not enter prompts or transition probabilities.

D.3 Reconstruction and technical provenance details

The frozen execution-source digest inadvertently enumerated one ignored 224-byte Python bytecode cache outside a conventional `__pycache__` directory. The cache was not scientific source and is not distributed as a repository artifact. A versioned provenance audit reconstructs the legacy digest from its recorded path and hash and additionally reports a reproducible cache-free semantic-source digest. This correction changes neither the protocol nor any formal trajectory or result.

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